RS-DGC: Exploring Neighborhood Statistics for Dynamic Gradient Compression on Remote Sensing Image Interpretation

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***Abstract*— Distributed deep learning has recently been attracting more attention in remote sensing (RS) applications due to the challenges posed by the increased amount of open data that are produced daily by Earth observation programs. However, the high communication costs of sending model updates among multiple nodes are a significant bottleneck for scalable distributed learning. Gradient sparsification has been validated as an effective gradient compression (GC) technique for reducing communication costs and thus accelerating the training speed. Existing state-of-the-art gradient sparsification methods are mostly based on the "larger-absolute-more-important" criterion, ignoring the importance of small gradients, which is generally observed to affect the performance. Inspired by informative representation of manifold structures from neighborhood information, we propose a simple yet effective dynamic GC scheme leveraging neighborhood statistics indicator for RS image interpretation, termed RS-DGC. We first enhance the interdependence between gradients by introducing the gradient neighborhood to reduce the effect of random noise. The key component of RS-DGC is a neighborhood statistical indicator (NSI), which can quantify the importance of gradients within a specified neighborhood on each node to sparsify the local gradients before gradient transmission in each iteration. Further, a layer-wise dynamic compression scheme is proposed to track the important changes of each layer in real time. Extensive downstream tasks validate the superiority of our method in terms of intelligent interpretation of RS images. For example, we achieve an accuracy improvement of 0.51% with more than 50× communication compression on the NWPU-RESISC45 dataset using VGG-19 network. To the best of our knowledge, this is the first GC method designed for RS images and downstream tasks, achieving a successful trade-off between high compression ratio and performance.**

***Index Terms*— Deep learning, distributed learning, gradient compression (GC), neighborhood information, remote sensing (RS).**

I. Introduction

With the rapid advancements in satellite technology and sensor technology, researchers can now effortlessly gather a vast amount of high-quality remote sensing (RS) images. This makes it a perfect target area for data-driven applications. In recent years, developments in software and hardware technology have greatly influenced Earth observation applications, particularly in RS image interpretation and deduction. These advancements have facilitated access to datasets of superior quality, at faster acquisition rates [1]. These RS data can be used for downstream tasks such as object detection, image classification, and land cover classification, and are of great significance in urban planning, agriculture, environmental monitoring, and other fields [2], [3], [4], [5]. As a method to accelerate deep learning, distributed learning [6] can further help us solve the problem of intelligent interpretation of RS data efficiently, especially in the age of RS big data [7]. Consequently, significant efforts have been made to develop a multitraining nodes constellation system aimed at obtaining more efficient and comprehensive results in the analysis of RS data.

However, the communication costs keep rising with more and more nodes. This leads to notable delays in transmission, significantly affecting the training speed within a satellite distributed platform that is already constrained by limited resources [8].

In recent years, the swift progress of deep neural networks (DNNs) has provided technical support for a wide range of visual tasks in the field of RS [9], [10], [11], [12], [13]. Its powerful image processing capabilities have been successfully used in complex scene interpretation. For example, Gao et al. [14] proposed a wetland classification method based on deep feature interaction network (DFINet). They designed a deep cross attention module to extract auto-correlation and cross correlation information from multisource feature pairs, emphasizing complementary information that is meaningful for classification. However, with the increasing complexity and fineness requirement of target recognition in RS tasks, the scale of deep learning models is also increasing exponentially, leading to a large amount of redundancy in parameter transmission between nodes. More seriously, the high communication costs required for over-parameterization hinders the application of RS technology in scenarios with limited physical space.

Gradient compression (GC) has emerged as an effective approach to reduce communication costs during distributed learning. Many effective techniques have been proposed to

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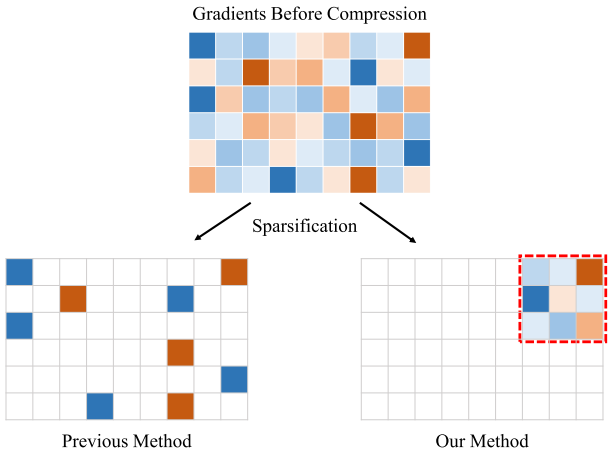


Fig. 1. Illustration of the compressing criterion for absolute-based approach and the proposed method. We interpret a layer's gradients as a matrix. Coordinate values are color coded (positive, negative), where deeper color denotes larger absolute value of the gradient. For the previous criterion, only the gradients with the largest absolute value are kept. In contrast, RS-DGC selects gradient neighborhoods that contribute more to network updating.

compress gradients, such as gradient sparsification [15], [16], [17], [18], quantization [19], [20], [21], [22], [23], [24], and low-rank decomposition [25], [26], [27], [28]. Among them, gradient sparsification transmits only a fraction of the gradient elements for full ones without compromising accuracy, thus offering a higher compression ratio (e.g., 0.1% in DGC [29]) compared to gradient quantization and low-rank decomposition. DGC introduced local gradient accumulation and momentum correction, reducing the communication costs in a more aggressive way. Nevertheless, DGC ignored the significance of small gradients, resulting in a certain degree of accuracy loss.

Current practice [30], [31] performs GC by following the "larger-absolute-more-important" criterion, which believes that gradients with smaller absolute values can be sparsed safely due to their less importance. As shown in Fig. 1, after calculating the absolute values of gradients in a model, a pre-specified or pre-computed threshold v is utilized to select gradients whose absolute value is smaller than it. However, this criterion ignores the effect of small gradients on model updating, leading to some relatively important neurons being unable to be updated for a long time. From a network training perspective, although large gradients can be effective in model updating, they often lead to challenging stability of convergency that degrades the model's performance. Furthermore, most existing GC methods treat each gradient as an independent individual and evaluate its importance. Yet, they ignore the influence of neighborhood information. Gaining contextual information from the neighborhood could learn informative structure-aware node representations [32]. The gradient's neighborhood information encompasses nearby gradient direction and magnitude, offering valuable insights for GC and reconstruction. Leveraging this information enhances our comprehension of the gradient's distribution and trends, leading to minimizing distortion in the compressed gradients.

Despite the possibility of noise or inaccuracies, it is crucial to emphasize that small gradients should be neither disregarded nor replaced. These seemingly diminutive gradients, despite their small magnitudes, harbor essential information vital to the training process. They actively contribute to the convergence of models and significantly influence achieving optimal performance. Previous studies have also pointed out that we should take the exploration of gradient information (i.e., sampling small gradients) into account [33], [34], [35]. Employing these smaller gradients reasonably can enhance model stability and generalization, providing a safeguard against overfitting and local minimum challenges.

To address these challenges, we propose RS-DGC, a dynamic GC method, by exploring the neighborhood statistics information for RS image interpretation. Some high-dimensional data, especially RS data, are actually a low-dimensional manifold structure embedded in a high-dimensional space [36]. Inspired by this, we introduce neighborhood statistical indicator (NSI), a simple yet powerful indicator, to obtain the sparse representation of raw gradients efficiently. Different from the previous methods, which sparsed gradients with small absolute, RS-DGC chooses the gradients with the most information. Specifically, we calculate the NSI by computing the mean and standard deviation of gradients within a specified neighborhood. Employing the NSI allows us to quantitatively assess gradient importance, capturing both relatively small but significant gradients and larger ones. This strategy ensures that the gradients that need to be transmitted offer richer information and are more conducive to model updating. Yet, due to the dynamic nature of DNNs training, the importance of gradients across various layers fluctuates throughout the process. A fixed compression ratio fails to dynamically monitor the evolving significance of each layer, resulting in a certain degree of accuracy compromise. Consequently, a real-time dynamic compress policy that can adapt to the dynamic convergence process must be developed.

We thoroughly evaluated the effectiveness and generalization of our proposed method by performing extensive experiments on two downstream tasks of RS. These tasks include RS image classification and HSI/LiDAR multimodal scene classification. To assess the performance of our method, we employ various neural networks such as convolutional neural networks (CNNs) and vision transformer (ViT), and conduct comparative experiments against state-of-the-art algorithms. The experimental results indicate that our approach exhibits competitiveness, particularly in scenarios with high sparsity compared to existing GC methods.

The major contributions of our work are outlined as follows.

1) We propose a new GC criterion called NSI. To our knowledge, it is the first time that taking advantage of gradient neighborhood to quantify the importance of gradients rather than individual gradients. NSI eliminates the contingency of individual gradients to model update by obtaining contextual information and captures the most informative gradient through gradient statistical indicators, which enables a more stable compressing.
2) We propose a layer-wise dynamic adaptive compression ratio adjustment strategy to track the important changes of each layer. It can adaptively adjust the compression ratio of each layer in the process of model training

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with weight information. This method helps not only to enhance model accuracy, but also to maintain the compression ratio at a very high level.

3) Extensive experiments on multiple neural networks and RS downstream tasks are conducted to show the effectiveness of our proposed RS-DGC. Experimental results show that our method is superior to mainstream techniques in terms of accuracy and communication costs.

The rest of this article is organized as follows. In Section II, we introduce the related works. In Section III, we describe the proposed approach in detail. In Section IV, a series of experiments on several downstream tasks are reported. Finally, the conclusions of the study are drawn in Section V.

II. Related Works

A. Gradient Compression

Broadly speaking, GC methods can be roughly divided into three main categories: gradient quantization, low-rank decomposition, and gradient sparsification. We will introduce them, respectively.

1) Quantization: Quantization methods reduce the number of bits of each element of the gradients by performing a low bit representation of float32 data of the original gradient. SignSGD [20] quantizes each float32 element of the gradient to 1-bit sign, by mapping the negative components to -1 and the others to $+1$. Seide et al. [37] proposed an error compensation scheme to reduce the error caused by quantization. Each time, the quantization error of the previous gradient is added to the current gradient and then quantized. Q-SGD [19] quantizes each element via randomized rounding. Wu et al. [23] proposed ECQ-SGD, which also utilized the error compensation method. Unlike 1-bit SGD, ECQ-SGD incorporated the accumulation of all previous quantization errors as the compensation error. However, quantization can at most reduce the communication costs by $32\times$. And because the low-bit representation of gradients will lose information, it is almost impossible to avoid accuracy loss.

2) Low-Rank Decomposition: Low-rank decomposition pertains to the process of decomposing the original gradient tensor into multiple matrices or tensors of lower rank. Wang et al. [26] employed computationally intensive singular value decomposition (SVD) to accomplish low-rank decomposition, whereas Vogels et al. [27] utilized power iteration to calculate lower-rank matrices with significantly reduced computational costs for decomposition. However, the compression ratio of low-rank decomposition is also very limited.

3) Sparsification: Gradient sparsification is a highly promising approach for reducing communication overhead compared to the aforementioned methods. By leveraging sparsity, this method effectively lowers communication costs by transmitting only a small subset of gradients. Random-k [38] sparsifiers randomly select k elements from the gradients for communication, whereas Top-k [39] sparsifiers select elements with higher magnitudes by sorting the gradients for transmission. Compared to Top-k sparsifiers, random-k sparsifiers exhibit higher randomness, which could not guarantee the selection of larger gradients. Besides, there are many other ideas for gradient sparsification. Gradients between different nodes have high mutual information, Abrahamyan et al. [40] took advantage of inter-node redundancy to improve compression efficiency by training auto-encoder (AE) to compress gradient information and decompress it on other nodes. Tao et al. [41] proposed CE-SGD, which could adaptively adjust the gradient sparsity according to the model's feedback. It also selectively transmitted the gradients based on their degree of participation in the back propagation.

B. RS Distributed Learning

The explosive growth of RS data presents significant challenges for effectively processing data to support various RS applications [42], [43], [44], [45]. With the increasing volume and speed of data generation in RS, processing has inevitably become a big data problem [46], [47], [48]. As a result, numerous recent studies have concentrated on the distributed parallel processing of RS big data. Boulila et al. [49] proposed RS-DCNN, an RS large datasets classification method based on distributed deep learning, which performs data parallel training on the Apache Spark framework. Martel et al. [50] proposed a parallel algorithm called pFUN for full unmixing of remotely sensed hyperspectral images on two GPUs by using CUDA programming and the cuBLAS library, respectively. Certain computation-intensive operations in the unmixing flow are performed on these two GPUs to enhance the computational efficiency of the entire processing flow. Sun et al. [48] proposed a new big data framework to process large amounts of RS images on cloud computing platforms. In addition to leveraging the parallel processing capabilities of cloud computing to deal with large-scale RS data, this framework also incorporates task scheduling strategies to further exploit the parallelism of the distributed processing stage. Li et al. [51] proposed FedFusion, a manifold-driven multimodal fusion framework, and developed a multimodal federated learning module that achieves iterative updates to the network parameters of each client through global weighted averaging. Additionally, communication costs among clients are compressed through low-rank decomposition. Fadlullah and Kato [52] proposed an asynchronous update federated learning model for edge nodes to build local artificial intelligence models for intelligent RS, incorporating the use case of forest fire detection without explicit data exchange with the cloud. This protects data privacy while reducing network overhead. Xu and Mao [53] proposed a federated learning approach to identify vehicle targets in RS images, addressing data privacy concerns during RS data training. Zhu et al. [54] proposed a blockchain-based PPFL RS image classification framework that can withstand model poisoning attacks without compromising user privacy. They first introduced the proof of accuracy (PoA) method to provide a trusted evaluation of encrypted models. Then, they designed a security aggregation framework based on PoA, which achieves robustness in most adversarial environments. However, most of the current distributed research in RS focuses more on privacy protection and lacks improvement and enhancement in communication efficiency among nodes.

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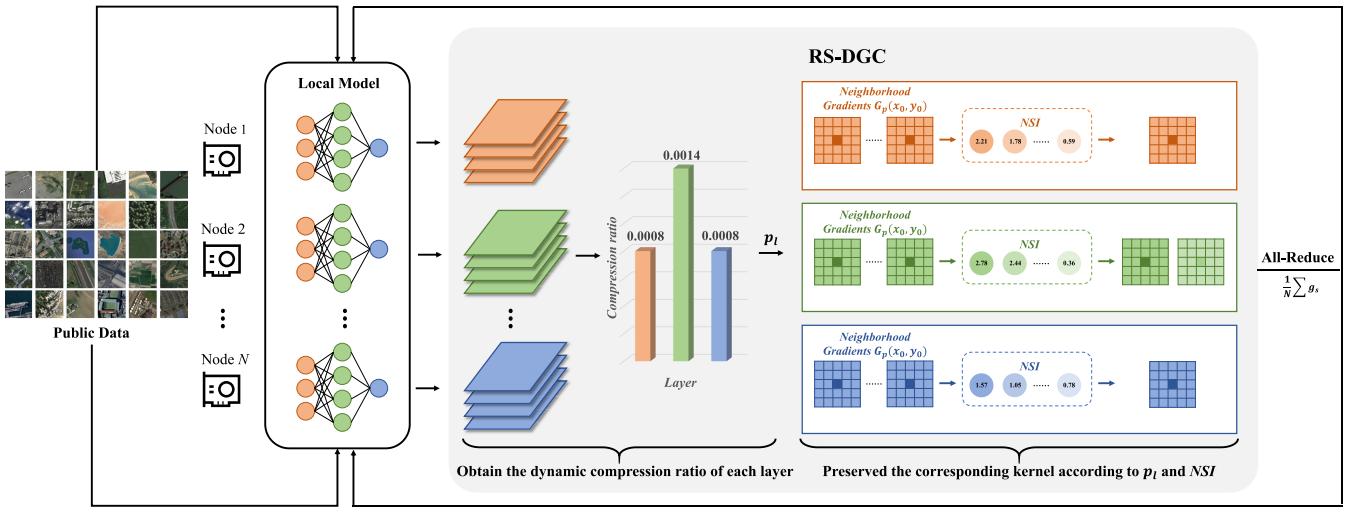


Fig. 2. Main workflow of RS-DGC. Firstly, utilizing the dynamic compression ratio strategy, each node calculates the compression ratio for every layer of the local network. Then, each node undertakes independent local training using its own data. After the backward propagation, GC is performed based on the NSI of each neighborhood and the corresponding compression ratio corresponding to this layer.

III. Methodology

This section describes the formulation of our GC method. First, we introduce the formulation of our problem. Then, we describe the dynamic compression strategy. Finally, we introduce the GC method based on NSI.

A. Problem Definition

A distributed learning system consists of N nodes, and each node can perform local training based on RS dataset D_n $(1 \leq n \leq N)$. The purpose of GC is to reduce the gradient elements that need to be transmitted as much as possible without losing the accuracy of the model, thereby minimizing the communication costs. Our aim is to solve the following optimization problem:

$$\begin{aligned} &\min \sum_{i=1}^{N} C(\Phi(\boldsymbol{g_i}, p)) \\ &\text{s.t. } \mathcal{L}_{\text{SGD}}(\boldsymbol{g_i}, D_i) \geq \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{S\text{-SGD}}\left(\Phi\left(\boldsymbol{g_i}, p_i^j\right), D_i\right) \end{aligned} \tag{1}$$

where $\mathcal{C}(\cdot)$ represents communication cost, $\Phi(\cdot)$ represents the operation to the gradients of each node. $\mathcal{L}_{\text{SGD}}$ is the loss function using stochastic gradient descent (SGD), and $\mathcal{L}_{S\text{-SGD}}$ is the loss function after GC. g_n is gradients at node n, p is the global compression ratio.

The gradient sparsification and synchronization using classic data-parallel SGD can be formulated as

$$\begin{aligned} &\boldsymbol{g_t} = \nabla f_n(\mathbf{X_t}; D_n), \quad \boldsymbol{g_t^s} = \text{Sparse}(\boldsymbol{g_t}) \\ &\boldsymbol{g_t^*} = \text{Sync}\left(\boldsymbol{g_t^s}\right), \quad \mathbf{X_{t+1}} = \mathbf{X_t} - \eta \boldsymbol{g_t^*}. \end{aligned} \tag{2}$$

At each iteration t, $\mathbf{X_t}$ is the model parameters, $\boldsymbol{g_t}$ is the local raw gradients, $\boldsymbol{g_t^s}$ is the local gradients after sparsification, and $\boldsymbol{g_t^*}$ is the global gradients after synchronization. η represents the learning rate. The Sparse$(\cdot)$ function is employed to sparsify the original gradient, thereby setting zeros for most gradient values while retaining important gradients for updates. Sync$(\cdot)$ function denotes the synchronization of gradient information across nodes, typically achieved through methods such as all-reduce, ring all-reduce, and others.

B. Dynamic Compression Strategy

When considering DNNs, it is crucial to acknowledge that the significance of each layer within the model is not uniform. We posit that layers with larger weight parameters hold greater importance. This proposition is primarily rooted in the fact that weight parameters dictate the extent to which each layer processes input data. By employing larger weight values, the input data can exert a more pronounced influence on subsequent layers, thereby amplifying the importance of the given layer. Conversely, smaller weight values attenuate the impact of input data, thereby diminishing the layer's contribution to the overall network. Furthermore, the utilization of larger weight values may result in the propagation of more substantial gradient values throughout that layer during the backward propagation. Consequently, it becomes necessary to determine distinct compression rates for individual layers. Additionally, the importance of each layer can experience variations during the neural network training process. Consequently, we derive real-time dynamic compression rates for each layer, which are subsequently employed in subsequent sparsification procedures.

Fig. 3 illustrates our policy for dynamic GC. Specifically, we assess the importance of each layer based on the magnitude of its weight, that is,

$$\mu_l^i = \left|w_l^i\right| \tag{3}$$

where w_l^i is the ith weight of layer l, $|\cdot|$ returns the absolute value of its input. We then conduct a cross-layer ranking across the whole network by the numerical order of μ_l^i. Assuming a global compression ratio p, the compression ratio of each layer can be easily determined by sparsifying the weights with the

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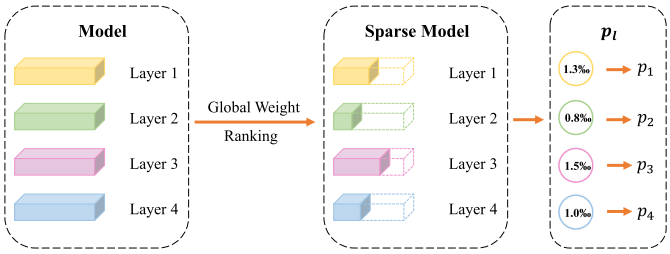


Fig. 3. Framework of dynamic GC method. We assess neuron significance based on the absolute magnitude of weights, postulating that layers with greater average weights hold more significance. The sparsity of each layer within the model is established through cross-layer global importance ranking, determining the layer's dynamic compression ratio.

lowest ranks. As a result, each weight w_l^i will be transformed into a sparse representation $\widehat{w}_l^i$ whose elements are defined as

$$\widehat{w}_l^i = \begin{cases} 0, & \mu_l^i \text{ is among the lowest-rated} \\ w_l^i, & \text{otherwise.} \end{cases} \tag{4}$$

Previous work has indicated that more than 99% of gradients can be safely removed by gradient sparsity with little impact on training accuracy [29]. We further consider the importance ranking of different layers of the network, so that the global redundant information can be tracked more accurately, which provides useful information for the subsequent GC.

A straightforward method for quantifying the importance of each layer of the network is to determine per-layer compression rate p_l based on the per-layer sparsity after above operation as

$$p_l = \frac{\sum_i \delta\left(\widehat{w}_l^i \neq 0\right)}{G_l} \tag{5}$$

where G_l is the total number of gradients in lth layer, and $\delta(\cdot)$ is an indicator function, which returns 1 if the input is true, and 0 otherwise. The definition of p_l in (5) equalizes the sparsity of the lth layer. To optimize training efficiency, we refrain from executing the aforementioned operation in every iteration. Instead, we determine the frequency of implementing the dynamic compression ratio algorithm based on several factors such as computing resources, efficiency requirements, downstream tasks, and other relevant considerations.

C. Neighborhood Statistics Indicator (NSI)

In order to effectively prioritize gradients for transmission, we introduce NSI as a quantifiable measure to assess the significance of gradients. Most existing gradient GC measure the importance of gradients by each individual gradient itself. Differently, we consider a gradient and its neighborhood as a whole.

Generally, a convolution layer transforms an input tensor $\boldsymbol{\mathcal{X}} \in \mathcal{R}^{C\times H_{\text{in}}\times W_{\text{in}}}$ into an output tensor $\boldsymbol{\mathcal{Y}} \in \mathcal{R}^{N\times H_{\text{out}}\times W_{\text{out}}}$ by using the filters $\boldsymbol{\mathcal{W}} \in \mathcal{R}^{N\times C\times K_h\times K_w}$. Here, C is the number of the input feature maps, N is the number of the filters, and K_h and K_w are the height and width of a filter, respectively. Since weights and gradients have a one-to-one correspondence, the gradient shape of each convolution layer is consistent with the weight, that is, $\boldsymbol{g} \in \mathcal{R}^{N\times C\times K_h\times K_w}$. We convert the gradients of each layer into tensors on a 2-D plane. Let g_0 be a 2-D spatial plane coordinate (x_0, y_0). The Chebyshev distance between g_0 and (x, y) on the 2-D plane is defined as

$$D_{\text{Chebyshev}} = \max(|x - x_0|, |y - y_0|). \tag{6}$$

The neighboring gradients $\boldsymbol{G_p}(x_0, y_0)$ defined by Chebyshev distance is a set of gradients that are within $p \times p$ patch including (x_0, y_0) itself and is defined as follows:

$$\boldsymbol{G_p}(x_0, y_0) = \left\{ \boldsymbol{G}(x, y) \middle| |x - x_0| \leq \frac{p}{2}, |y - y_0| \leq \frac{p}{2} \right\} \tag{7}$$

where $\boldsymbol{G}(x, y)$ is the gradient at position (x, y) in layer l. We usually regard the gradient value corresponding to a convolution kernel as a whole. For example, for a 3×3 convolution kernel, we usually set $p = 3$. This is because in DNNs, each neuron is independent, and treating it as a whole can effectively measure the importance of each neuron, rather than treating each gradient value in isolation. The neighborhood defined by Chebyshev distance is shown in Fig. 1, where $p = 3$.

We measure the importance of gradients in neighborhood $\boldsymbol{G_p}(x_0, y_0)$ by calculating the NSI

$$\text{NSI}_p(x_0, y_0) = \alpha \times \text{mean}_p(x_0, y_0) + (1 - \alpha) \times \text{std}_p(x_0, y_0) \tag{8}$$

where α is the hyperparameter between 0 and 1, and it can be adjusted for different models. $\text{mean}_p(x_0, y_0)$ and $\text{std}_p(x_0, y_0)$ are the mean of absolute value and standard deviation of the gradients in the neighborhood $\boldsymbol{G_p}(x_0, y_0)$. It is worth noting that we have opted to utilize standard deviation instead of variance in our study. The main reason for this choice is that standard deviation and mean share the same unit. They are

Algorithm 1 Algorithm of RS-DGC

Input: The global compression ratio p, raw gradient $\boldsymbol{g}$, hyperparameter α.
Output: Sparse gradient $\boldsymbol{g_s}$.

for each epoch **do**
  $\mu_l^i = |w_l^i|$;
  Conduct a global ranking of weights by μ_l^i and obtain the sparse weights by removing the bottom-ranked weights via Eq.4;
  for $i = 1 \rightarrow L$ **do**
    Obtain the per-layer compression ratio p_l via Eq.5;
  end for
  for each iteration **do**
    $NSI_p^k(x_0, y_0) = \alpha \times mean_p^k(x_0, y_0) + (1 - \alpha) \times std_p^k(x_0, y_0)$;
    Conduct a ranking of $NSI_p^k(x_0, y_0)$;
    Obtain the sparse gradients $\boldsymbol{g_s}$ by preserving the gradients in neighborhood with the top-ranked NSI;
  end for
end for
return $\boldsymbol{g_s}$

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defined as follows:

$$\text{mean}_p(x_0, y_0) = \left\| \frac{1}{p^2} \sum_{i=1}^{p^2} |g_i| \right\|_2 \tag{9}$$

$$\text{std}_p(x_0, y_0) = \left\| \frac{1}{p^2} \sum_{i=1}^{p^2} \left(g_i - \frac{1}{p^2} \sum_{j=1}^{p^2} g_j \right)^2 \right\|_2^{\frac{1}{2}}. \tag{10}$$

Suppose we get the compression ratio corresponding to the nth node and the lth layer of the model p_n^l, and then we calculate the number of gradients that is need to be preserved

$$k_n^l = \left\lceil \frac{N \times C \times K_h \times K_w}{p \times p} \times p_n^l \right\rceil. \tag{11}$$

Then, we will preserve all gradient values in the neighborhood corresponding to the Top-k NSI value and transmit the corresponding neighborhood index.

Therefore, our approach can be seen as a variant of Top-k sparsity. It is worth noting that our method differs from Top-k sparsity in two ways.

1) By introducing gradient neighborhood, we aggregate all the gradients within the specified neighborhood as a unit to mitigate the contingency caused by individual gradients. This approach effectively minimizes the influence of random noise, resulting in smoother and more stable gradients.
2) Statistic indicators, particularly the standard deviation, assist us in capturing significant yet relatively small gradients; thereby, preventing the model from overfitting. Moreover, these small gradients encompass crucial fine-tuning information regarding the model's weight, playing a major role in sustaining optimal model performance.

To solve the problem of gradient optimization direction deviation caused by gradient accumulation directly, Lin et al. [29] proposed local gradient accumulation using momentum correction

$$\begin{aligned} \boldsymbol{u}_{k,t} &= m\boldsymbol{u}_{k,t-1} + \boldsymbol{g}_{k,t} \\ \boldsymbol{v}_{k,t} &= \boldsymbol{v}_{k,t-1} + \boldsymbol{u}_{k,t} \\ \boldsymbol{w}_{t+1} &= \boldsymbol{w}_t - \eta \sum_{k=1}^{N} \text{sparse}\left(\boldsymbol{v}_{k,t}\right) \end{aligned} \tag{12}$$

where the first two terms are the corrected local gradient accumulation, and the accumulation result $\boldsymbol{v}_{k,t}$ is used for the subsequent sparsification and communication. We have also adopted this idea.

IV. Experiments

In this section, we conduct extensive experiments on two classical downstream tasks in the RS field to evaluate the comprehensive performance of the proposed method. Specifically, it includes RS image classification and multimodal scene classification. We first present the datasets in Section IV-A. Then, we introduce the evaluation metrics and parameter setting in Section IV-B. In Section IV-C, we conduct ablation experiments. The effectiveness of each module in the proposed method is further verified. In Sections IV-D and IV-E, the performance of RS-DGC on the two downstream tasks is reported, respectively. In addition, a series of comparative experiments are carried out to compare with the existing mainstream methods. Finally, in Section IV-F, we show the actual communication costs reduction of our method.

A. Data Description

1) UC Merced Land-Use Dataset: The UC Merced land-use dataset (UCML-21) [55] was released in October 2010 by the Computer Vision Laboratory at the University of California, Merced. The dataset images are derived from manually extracting large images from the U.S. geological survey (USGS) national map urban area imagery collection. The image size is 256 × 256 pixels and the spatial resolution is 1 foot. A total of 21 categories including agricultural area, aircraft, and baseball field are included, and each category has 100 images. Fig. 4 shows some sample images randomly selected from UCML-21.

2) NWPU-RESISC45 Dataset: The NWPU-RESISC45 RS dataset (NWPU-45) [56] is a large-scale public dataset released by Northwestern Polytechnical University for the scene classification of RS images. It is extracted from Google Earth (Google Inc.) and covers more than 100 countries and regions in the world. This dataset contains 45 categories of scenes, and the scene samples of each category are displayed as shown in Fig. 5. Each category contains 700 images, and the size of each image is 256 × 256 pixels. Compared with UCML-21, it has the characteristics of large-scale and rich information contained in images.

3) HSI-LiDAR Houston2013 Dataset: The HSI-LiDAR Houston2013 dataset is a valuable Earth observation dataset that combines hyperspectral and LiDAR data for in-depth studies on land cover and environmental changes. This dataset was created by CosmiQ Works team and uses high-resolution WorldView-2 satellite imagery. Collected by the ITRES CASI-1500 imaging sensor in 2013, an extensive data collection campaign was conducted in the University of Houston campus area in Texas, USA. The hyperspectral component of this dataset comprises 144 bands, covering a wide wavelength range of 0.38–1.05 μm, with a spectral interval of 10 nm. On the other hand, the LiDAR dataset consists of a single-band image. The size of all images in the dataset is 349 × 1905 pixels, as shown in Fig. 6.

B. Evaluation Metrics and Parameter Setting

1) Evaluation Metrics: In RS image classification tasks, the network performance is measured using Top-1 accuracy. As for the multimodal scene classification task, we employ three indicators: overall accuracy (OA), average accuracy (AA), and the kappa coefficient. OA quantifies the proportion of correctly classified test samples out of the total number of test samples. AA represents the average accuracy across all categories. The kappa coefficient assesses the consistency between the classification map generated by the model and the provided true values.

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Fig. 4. Example images from the UCML-21 dataset containing 21 land-use scene categories: agricultural area, airplane, baseball diamond, beach, buildings, chaparral, dense residential area, forest, freeway, golf course, harbor, intersection, medium residential area, mobile homepark, overpass, parking lot, river, runway, sparse residential area, storage tanks, and tennis court.

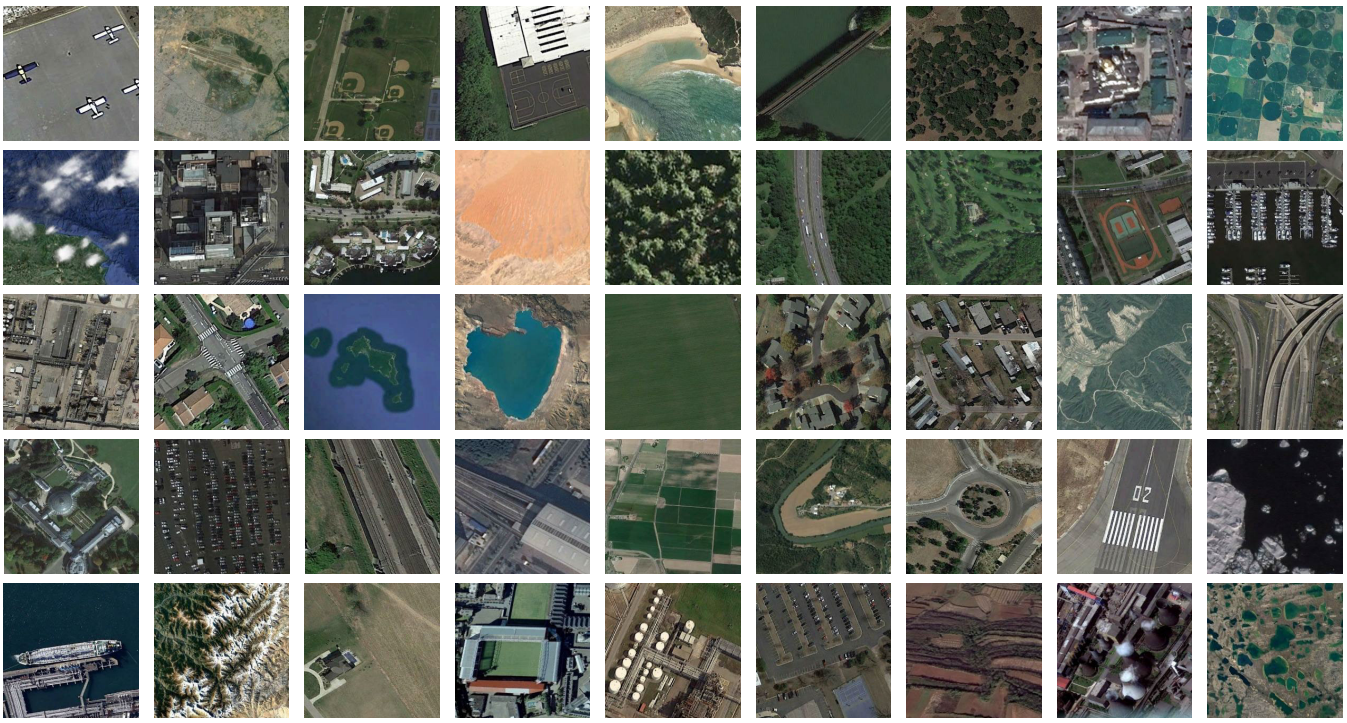


Fig. 5. Example images from the NWPU-45 dataset containing 45 land-use scene categories: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential area, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential area, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential area, stadium, storage tank, tennis court, terrace, thermal power station, and wetland.

2) Parameter Setting: The proposed GC method is implemented with PyTorch [57] on 4 NVIDIA RTX3090 GPUs with 24 GB VRAM, and we use NCCL as communication backend. To reduce the effect of random seeding, all results are the average of several experiments. The highest transmission rate that was achievable was 10 GB/s. For the RS image datasets UCML-21 and NWPU-45, the network is trained for 500 epochs. The initial learning rate is set to 0.1 and the batch size is 16. In the training phase, all the networks are optimized by SGD optimizer with a momentum of 0.9. The learning rate is updated by MultiStepLR strategy ($\gamma = 0.1$) when using VGGNet and CosineAnnealingLR when using ResNet. In the multimodal scene classification task, we set the training epoch to 300. Concretely, we adopt the Adam optimizer to update the network parameters with a batch size of 64. The learning rate is initially set as $5e^{-4}$ and then decayed by 0.9 every 20 steps,

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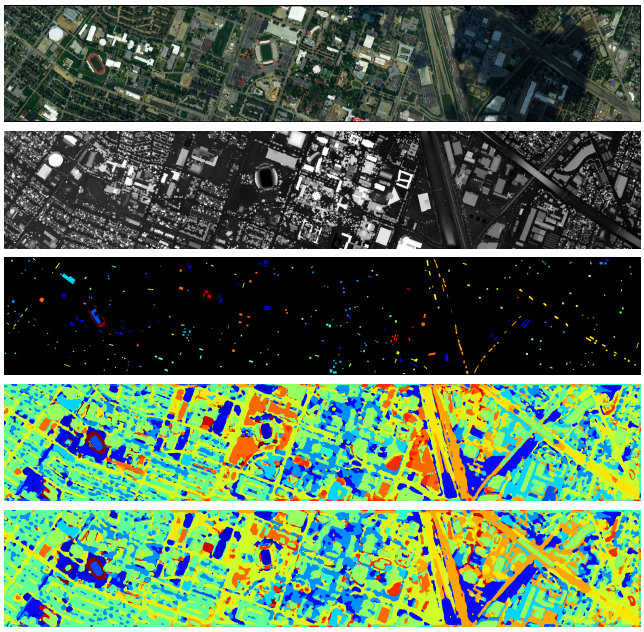


Fig. 6. Scene display and classification results visualization of the Houston2013 dataset. From top to bottom, the HSI of the Houston2013 dataset, the LiDAR image, the labels, the scene classification results using ExViT, and the classification results after RS-DGC.

TABLE I
ABLATION STUDY RESULTS FOR VGG-19 AND RESNET-56 ON UCML-21 DATASET

Models	Optimizations			Accuracy(%)	
	Neighbor	Statistic	Dynamic	Top-1 Acc.	Top-1 Acc.↑
VGG-19				87.96	
	✓			87.98	0.02
	✓	✓		88.03	0.05
	✓	✓	✓	88.57	0.54
ResNet-56				90.28	
	✓			90.31	0.03
	✓	✓		90.65	0.34
	✓	✓	✓	91.45	0.80

and weight decay is parameterized by 0 and $5e^{-4}$, respectively. The gradient sparsity is 99.9% (only 0.1% is non-zero), and the quantization bit width in QSGD is set to 1 bit.

C. Ablation Study

We conduct ablation experiments using the UCML-21 dataset. Networks used include VGG-19 [58] and ResNet-56 [59]. The results are shown in Table I.

1) Neighborhood Information: We first verified the importance of neighborhood gradient information. By comparing the first and second rows of VGG-19 and ResNet-56 in Table I, we can see that neighborhood gradients boost VGG-19 and ResNet-56 by 0.02% and 0.03%, respectively.

2) Statistic Indicator: To verify the effectiveness of statistic indicator, we compared the Top-1 accuracy on UCML-21 datasets with or without standard deviation. As shown in Table I, NSI helps improve VGG-19 and ResNet-56 by 0.05% and 0.34%.

3) Dynamic Compression: Our third experiment combines fixed compression ratio and our dynamic compress strategy on VGG-19 and ResNet-56. The fixed compression ratio is $p = 0.001$, and we also set the global compression ratio as 0.001 in our dynamic compress strategy. As Table I lists, VGG-19 and ResNet-56 obtain 0.54% and 0.80% accuracy improvement over fixed compression ratio, respectively, showing that the dynamic compress strategy is effective.

TABLE II
TOP-1 ACCURACY AND GRADIENT SPARSIFICATION RATIO ON UCML-21 DATASET

Models	Methods	Top-1 Acc.(%)		Sparsification Ratio
VGG-19	SGD	92.38		1×
	Top-k [39]	86.01	-6.37	1000×
	DGC [29]	86.19	-6.19	1000×
	QSGD [19]	87.27	-5.11	32×
	RS-DGC	**88.57**	**-3.81**	1000×
ResNet-56	SGD	95.24		1×
	Top-k [39]	90.89	-4.35	1000×
	DGC [29]	91.43	-3.81	1000×
	QSGD [19]	90.77	-4.47	32×
	RS-DGC	**91.45**	**-3.79**	1000×
ResNet-110	SGD	95.71		1×
	Top-k [39]	93.10	-2.61	1000×
	DGC [29]	93.33	-2.38	1000×
	QSGD [19]	94.79	-0.92	32×
	RS-DGC	**95.24**	**-0.47**	1000×

D. RS Image Classification

We conduct the RS image classification experiment on UCML-21 and NWPU-45 datasets, with various backbones including VGG-19, ResNet-56, and ResNet-110. We choose the results of distributed SGD optimization as the baseline.

In the experiment of UCML-21, Table II shows the Top-1 accuracy and gradient sparsification ratio compared with other methods. As an improvement of the Top-k method, DGC has indeed improved model performance. However, DGC also has a large accuracy loss on RS image classification. For example, the accuracy loss of VGG-19 is 6.19%, which is not allowed in practice. In contrast, the accuracy loss of our proposed method is only 3.81% and 3.79% when using VGG-19 and ResNet-56, respectively. The classification results on NWPU-45 dataset are shown in Table III. Compared to other comparison methods, our method minimizes accuracy loss and even achieves an accuracy improvement of 0.51% when training the NWPU-45 dataset on the VGG-19 network. One possible explanation we can deduce is that neural networks are overparameterized and their gradient

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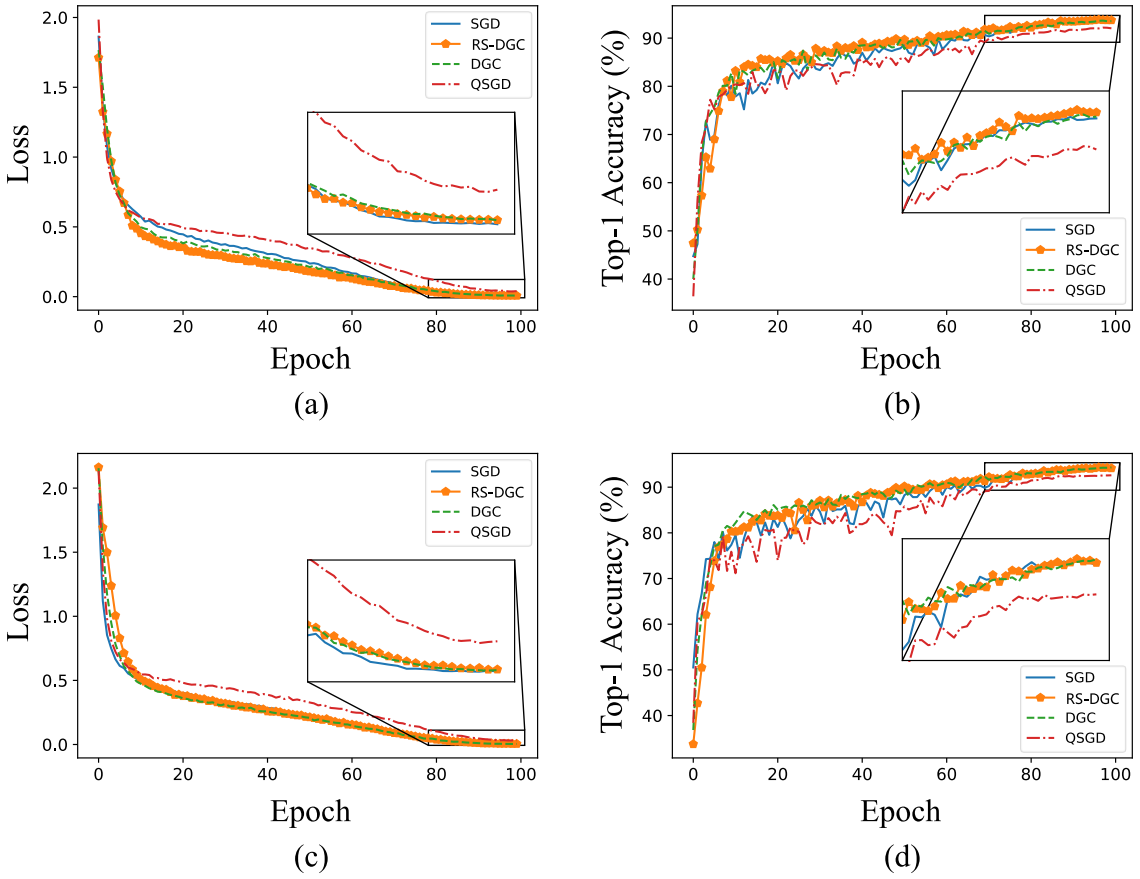


Fig. 7. Learning curves of ResNet-110 and VGG-19 on UCML-21 and NWPU-45 datasets. (a) Training loss of ResNet-110 on UCML-21. (b) Top-1 accuracy of ResNet-110 on UCML-21. (c) Training loss of VGG-19 on NWPU-45. (d) Top-1 accuracy of VGG-19 on NWPU-45.

TABLE III
TOP-1 ACCURACY ON NWPU-45 DATASET

Dataset	**NWPU-45**		
Method	**VGG-19**	**ResNet-56**	**ResNet-110**
SGD	94.06	91.08	86.29
Top-k [39]	93.88	87.11	83.17
DGC [29]	94.44	**87.27**	83.68
QSGD [19]	90.18	85.99	82.16
RS-DGC	**94.57**	87.21	**84.69**

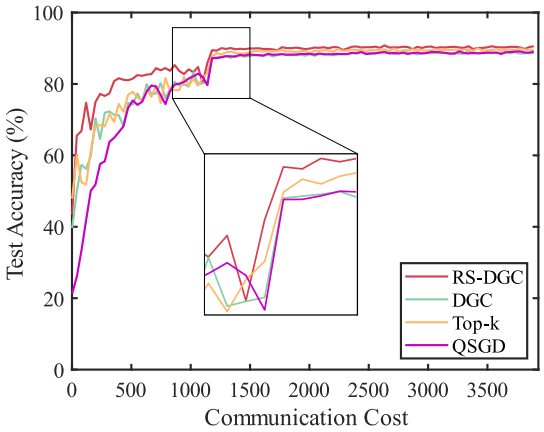


Fig. 8. Comparison of the communication costs for different methods on VGG-19 training on UCML-21 dataset.

information can be compressed to prevent overfitting of the model for more efficient updates. In addition, through the comparison of experimental results between ResNet-56 and ResNet-110 on the two datasets, we observe a gradual decrease in accuracy loss caused by the same GC ratio as the model size increased. This finding indicates that as the model size grows, the presence of redundant information within the model also increases, thereby reducing the impact of compressing the gradient information. Hence, it can be inferred that larger models exhibit a higher tolerance toward GC due to the increased redundancy within them.

Taking VGG-19 and ResNet-110 as examples, we demonstrate the evolution of training loss and Top-1 accuracy in Fig. 7. The learning curve of QSGD is worse than the baseline due to information loss of low-bit representation. It is worth noting that the training loss of RS-DGC decreases faster than other methods. Although the convergence speed of RS-DGC is slower than SGD and DGC at the beginning of training, it converges more rapidly after ten epochs. As the training process deepens, the stability of our method is better than SGD, and the final training accuracy is better than DGC.

E. Multimodal Scene Classification

To ensure a comprehensive verification of the versatility of our proposed method, we conducted experiments focusing on

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TABLE IV
OA, AA, AND KAPPA ON THE HSI-LIDAR HOUSTON2013 DATASET

Dataset	**The HSI-LiDAR Houston2013 dataset**					
Network	**CCR-Net** [60]			**ExViT** [61]		
Method	**OA**(%)	**AA**(%)	**Kappa**(%)	**OA**(%)	**AA**(%)	**Kappa**(%)
SGD	93.15	93.16	92.56	91.40	92.60	90.66
Top-k [39]	88.02	87.99	87.11	82.14	82.27	81.13
DGC [29]	90.11	**90.21**	89.53	84.96	86.67	85.72
QSGD [19]	88.49	88.57	87.12	81.16	81.89	80.11
RS-DGC	**90.17**	90.19	**89.59**	**85.99**	**86.71**	**85.78**

TABLE V
GRADIENT SIZE AND COMPRESSION RATIO PER EPOCH FOR TRAINING VGG-16 AND VGG-19 NETWORKS

Network	**VGG-16**		**VGG-19**	
Method	**Gradient Size(MB)**	**Compression Ratio**	**Gradient Size(MB)**	**Compression Ratio**
SGD	57.19	1 ×	77.45	1 ×
Top-k [39]	4.23	13×	5.18	15×
DGC [29]	1.19	48×	1.31	59×
RS-DGC	**1.19**	**48×**	**1.31**	**59×**

RS multimodal scene classification using two advanced multimodal fusion networks, namely CCR-Net [60] and ExViT [61], and the results are shown in Table IV. CCR-Net is a network architecture based on convolutional layers. After GC, the accuracy loss is small. Compared with other methods, our proposed method achieves minimal OA and kappa losses of 2.98% and 2.97%, respectively. For the transformer-based network architecture ExViT, the accuracy loss of various GC methods is relatively large. This may be due to the fact that the self-attention mechanism layer in ViT involves interactions across all locations. However, when compared with other methods, the method we proposed is still better, with OA, AA, and kappa reaching 85.99%, 86.71%, and 85.78%, respectively. Additionally, we present visualizations of the scene classification results obtained using ExViT. As demonstrated in Fig. 6, by comparing the fourth and fifth figures, despite GC, favorable classification outcomes are still attained.

F. Communication Costs

The communication cost and accuracy curves are shown in Fig. 8. As we can see, our method has the lowest communication overhead in achieving the same accuracy. In addition, we also calculated the actual gradient size that needs to be transmitted in each epoch when training the VGG-16 and VGG-19 networks. As shown in Table V, our method achieves more than 50× reduction in communication overhead compared to the baseline. At the same time, compared with DGC, our method achieves lower accuracy loss at the same compression ratio.

V. CONCLUSION

In this article, we propose a novel GC method for RS image interpretation, called RS-DGC, which resolves the contradiction between model performance and high compression ratio with gradient neighborhood and dynamic compression. In particular, we employ NSI to reasonably measure the importance of gradients. Considering the dynamic nature of DNNs training, a dynamic compression strategy is proposed to further enhance the final accuracy. Extensive RS downstream experiments prove the effectiveness and generalization of the proposed RS-DGC.

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